

## ASSIGNMENT-05

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AIM :- Develop a responsive web design using Express Framework to perform CRUD operations & deploy with Node JS. use MongoDB.

### OBJECTIVE :-

- 1) Develop a Full-Stack Web Application.
- 2) Demonstrate Backend Development & Deployment Proficiency.

### THEORY :-

Q1) What is the role of Express.js as a web framework for Node.js?

Ans- Provides a lightweight, fast, & flexible web framework built on top of Node.js.

- Simplifies handling HTTP requests & responses.
- Supports routing
- Middleware support for request processing.
- Enables building RESTful APIs & full-stack applications efficiently.
- Reduces boilerplate code compared to using pure Node.js.



Q2) Explain the concept of CRUD operations in the context of a web application.

Ans - (CRUD) stands for Create, Read, Update, Delete - the four basic operations of persistent storage.

- Create :- Add new data.
- Read → Retrieve data.
- Update :- Modify existing data.
- Delete :- Remove data.

Q3) Why is MongoDB a suitable choice for this project?

Ans - NoSQL, document-oriented database → stores data in flexible JSON-like format.

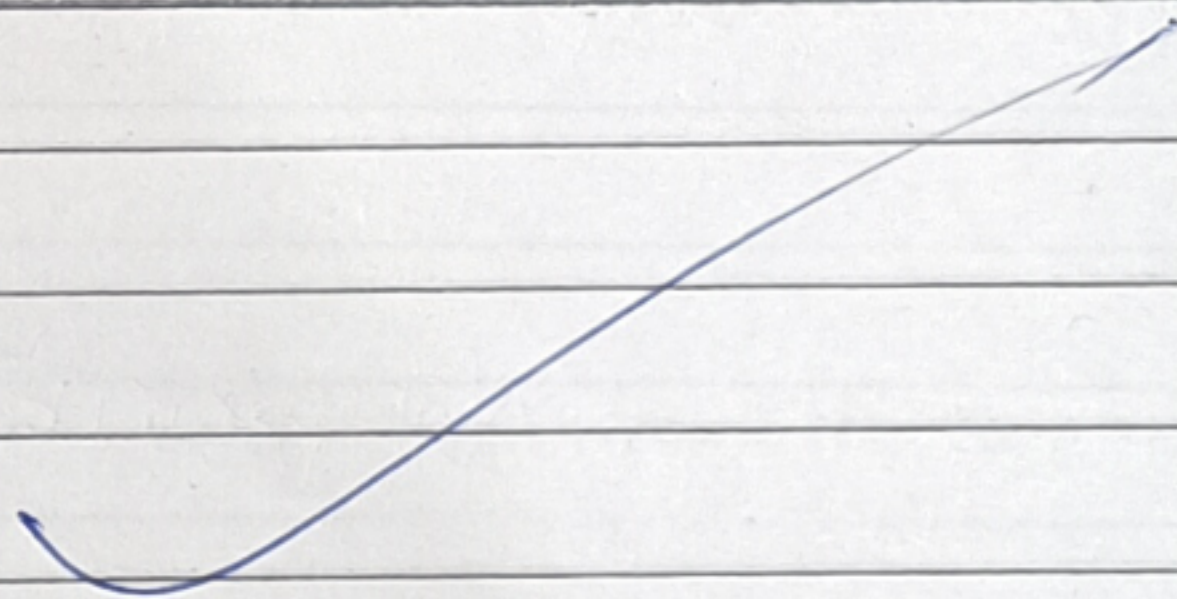
- Schema-less design → easy to adapt to evolving project requirement.
- Handles large-scale, unstructured, or semi-structured data efficiently.
- Strong integration with Node.js through the Mongoose library.
- Ideal for applications requiring real-time data & rapid development.



Q4. What steps are involved in deploying a Node.js & Express application?

Ans. Prepare app & set environment variables.

- Choose host (Heroku, AWS, etc.)
- Install Node.js & dependencies.
- Use PM2 for process management.
- Set up reverse proxy (NGINX) & SSL.
- Monitor & Maintain.



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